

SaaS B2B KPI EXECUTIVE DASHBOARD

Complete Case Study, Formula Guide & Interview Solution

■ PROJECT TYPE	
■ DOMAIN	SaaS B2B Collaboration Tool
■ DATA PERIOD	
■ BUSINESS MODEL	Land & Expand (Freemium → Enterprise)
■ CURRENT MRR	
■ NRR	124.3%
■■ CRITICAL ALERT	

This guide contains: Case Study → All KPI Formulas → Data Insights → Solutions → Interview Strategy

■ TABLE OF CONTENTS

- 1. EXECUTIVE SUMMARY — The Business Problem
- 2. KPI DEFINITIONS & FORMULAS — Every Metric Explained
- 3. THE THREE CRITICAL INSIGHTS — What The Data Reveals
- 4. SOLUTIONS & ACTION PLANS — From Insight to Execution
- 5. CHART-BY-CHART BREAKDOWN — How to Read Each Visual
- 6. INTERVIEW STRATEGY — How to Explain HTML vs Power BI
- 7. TECHNICAL DEEP DIVE — Stack, Architecture & Design Choices
- 8. APPENDIX — Raw Data Tables & Calculation Examples

1. EXECUTIVE SUMMARY

1.1 The Business Context

This dashboard analyzes a **SaaS B2B Collaboration Tool** operating on a **Land & Expand** business model. The product offers a freemium tier that converts users through a Product-Qualified Lead (PQL) funnel into paid Team, Business, and Enterprise plans. The company has reached **\$4.03M MRR** with healthy top-line growth, but underlying operational metrics reveal critical risks that threaten sustainable scaling.

1.2 The Paradox: Growth vs. Hidden Risk

At first glance, the business appears healthy: NRR of 124.3%, MRR growing 18% month-over-month, and gross margins at 88%. However, three critical insights emerge when we dig deeper into the data: **silent churn is masking revenue retention problems**, **feature adoption is bottlenecking expansion revenue**, and **a recent UI redesign is creating a support crisis that precedes churn by 90 days**.

1.3 Dashboard Purpose

This is not a reporting dashboard — it is a **decision-support system**. Every KPI connects to a causal insight, which connects to a recommended action with an assigned owner and measurable impact. The goal is to transform data observation into business action.

METRIC	CURRENT VALUE	TREND	STATUS	RISK LEVEL
Monthly Recurring Revenue	\$4.03M	↑ +18% MoM	■ Healthy	Low
Net Revenue Retention	124.3%	↑ +2.1pp vs Q1	■ Healthy	Medium
PQL Rate	5.6%	↑ +0.2pp MoM	■ Improving	Low
Time to Value	2.1 days	↓ -0.8 days	■ Improving	Low
Enterprise Sales Cycle	42 days	↓ -23 days	■ Improving	Low
Gross Margin	88%	↑ +1pp QoQ	■ Healthy	Low
Support Tickets / \$100K	12.1	↑ +25% spike	■ ALERT	HIGH
Silent Churn Rate	28.5%	↑ +12pp vs Jan	■ CRITICAL	CRITICAL

2. KPI DEFINITIONS & FORMULAS

Every metric on the dashboard follows a precise business definition. Below are the formulas, calculation logic, and interpretation guidelines for each KPI.

2.1 Monthly Recurring Revenue (MRR)

Definition: The predictable monthly revenue from all active subscriptions. This is the north star metric for any SaaS business.

Formula:

$$\text{MRR} = (\text{New MRR}) + (\text{Expansion MRR}) + (\text{Reactivation MRR}) - (\text{Contraction MRR}) - (\text{Churned MRR})$$

Component Breakdown:

- **New MRR:** Revenue from brand new customers in the month
- **Expansion MRR:** Additional revenue from existing customers (upgrades, add-ons, more seats)
- **Reactivation MRR:** Revenue from previously churned customers who returned
- **Contraction MRR:** Revenue lost from downgrades or seat reductions
- **Churned MRR:** Revenue lost from cancelled subscriptions

Current Dashboard Value: \$4.03M | **Growth:** +18% MoM

2.2 Net Revenue Retention (NRR)

Definition: Measures how much revenue you retain from existing customers over time, including expansion and churn. $\text{NRR} > 100\%$ means existing customers generate more revenue than they did at the start.

Formula:

$$\text{NRR} = [(\text{Starting MRR} + \text{Expansion MRR} - \text{Contraction MRR} - \text{Churned MRR}) / \text{Starting MRR}] \times 100$$

Benchmarks:

- **Best-in-class SaaS:** $\text{NRR} > 120\%$ (e.g., Snowflake ~169%, Datadog ~130%)
- **Good SaaS:** $\text{NRR} 100\text{-}120\%$
- **Warning zone:** $\text{NRR} < 100\%$ (you're shrinking even with new customers)

Current Dashboard Value: 124.3% | **Status:** Healthy but masking silent churn

2.3 Product-Qualified Lead (PQL) Rate

Definition: The percentage of free/trial users who demonstrate product value and are qualified as leads ready for sales engagement or self-serve conversion.

Formula:

$$\text{PQL Rate} = (\text{Number of PQLs} / \text{Total Free/Trial Users}) \times 100$$

PQL Criteria Example:

- Completed core action (e.g., sent 5+ messages)
- Invited 2+ team members
- Used 2+ features
- Active for 3+ consecutive days
- Reached a usage threshold (e.g., 100MB file storage)

Current Dashboard Value: 5.6% | Conversion to Paid: 72%

2.4 Time to Value (TTV)

Definition: The time it takes a new user to experience the core value proposition of the product. In SaaS, this is often called the 'Aha! Moment' — the point where the user understands why they need this tool.

Formula:

$$\text{TTV} = (\text{Timestamp of First Value Moment}) - (\text{Timestamp of Account Creation})$$

How to Define 'First Value Moment':

- For collaboration tools: First message sent AND first file shared
- For analytics tools: First report created and shared
- For CRM: First deal marked as won

Critical: The 'value moment' must be defined by the user, not the product team.

Current Dashboard Value: 2.1 days (avg) | Enterprise: 4.5 days

2.5 Enterprise Sales Cycle

Definition: The average number of days from first meaningful sales touch to closed-won deal for enterprise accounts.

Formula:

$$\text{Sales Cycle} = \Sigma(\text{Days from First Touch to Close}) / \text{Total Closed-Won Deals}$$

Segments to Track:

- **Self-serve (Team plan):** Typically 0-3 days (no sales touch)
- **Business plan:** 7-21 days (light sales touch)
- **Enterprise:** 30-90+ days (complex procurement, legal, security review)

Current Dashboard Value: 42 days | Win Rate: 40%

2.6 Gross Margin

Definition: The percentage of revenue retained after direct costs of delivering the service (COGS). In SaaS, this should be very high (70-90%+) because software has near-zero marginal cost.

Formula:

$$\text{Gross Margin} = [(\text{Total Revenue} - \text{COGS}) / \text{Total Revenue}] \times 100$$

SaaS COGS Includes:

- Cloud infrastructure (AWS/Azure/GCP hosting)
- Customer support costs (salaries, tools)
- Payment processing fees
- Third-party API costs
- **Excludes:** R&D, Sales & Marketing, G&A;

Current Dashboard Value: 88% | COGS Trend: Down 15% YoY

2.7 Support Tickets per \$100K MRR

Definition: A normalized metric that measures support efficiency relative to revenue scale. It answers: 'As we grow, are we getting more or less efficient at supporting customers?'

Formula:

$$\text{Support Tickets per \$100K} = (\text{Total Support Tickets in Period} / \text{MRR in \$100K units})$$

Example Calculation:

If MRR = \$4,030,000 and tickets = 487:
 MRR in \$100K units = \$4,030,000 / \$100,000 = 40.3
 Tickets per \$100K = 487 / 40.3 = **12.1 tickets per \$100K**

Why Normalize by Revenue?

Raw ticket volume will always grow as you add customers. Normalizing by revenue tells you if support is becoming a bigger or smaller problem **relative to your scale**.

Current Dashboard Value: 12.1 | Spike: +25% post-redesign

2.8 Silent Churn Rate

Definition: The percentage of paying accounts that show zero meaningful activity (logins, actions) for a defined period (typically 30-90 days) before officially cancelling. This is a **leading indicator** of future churn.

Formula:

$$\text{Silent Churn Rate} = (\text{Accounts with Zero Logins for 60+ Days} / \text{Total Paying Accounts}) \times 100$$

Why 'Silent' Churn is Dangerous:

- These accounts are still paying — so they inflate your retention numbers
- When they finally cancel, it's sudden and often in bulk (quarterly/annual renewals)
- They represent 'hidden' ARR at risk that doesn't show up in standard churn reports
- **Historical pattern:** 70% of silent churners had declining engagement for 6 weeks before going dark

Current Dashboard Value: 28.5% | At-Risk ARR: \$680K | Trend: ↑ +12pp since Jan

2.9 Feature Adoption Rate

Definition: The percentage of active users who have used a specific feature at least once within a given time period.

Formula:

$$\text{Feature Adoption Rate} = (\text{Users Who Used Feature} / \text{Total Active Users}) \times 100$$

Key Insight from Dashboard:

Accounts using **4+ features** generate **3x more expansion revenue** than single-feature users. Yet **70% of users never discover feature #3** (Integrations/Automation). This is a product discovery problem, not a product value problem.

2.10 PQL Funnel Conversion Rates

Funnel Stages & Formulas:

Stage 1 → Stage 2: PQL Rate = (PQLs / Free Users) × 100 = (750 / 13,500) × 100 = **5.6%**

Stage 2 → Stage 3: SQL Rate = (SQLs / PQLs) × 100 = (450 / 750) × 100 = **60.0%**

Stage 3 → Stage 4: Win Rate = (Closed Won / SQLs) × 100 = (280 / 450) × 100 = **62.2%**

Overall: Free → Paid = (280 / 13,500) × 100 = **2.07%**

3. THE THREE CRITICAL INSIGHTS

The dashboard reveals three interconnected business problems that, if unaddressed, will reverse the company's growth trajectory. Each insight follows the pattern: **Observation** → **Root Cause** → **Financial Impact** → **Recommended Action**.

■ INSIGHT 1: NRR is Masking a Silent Churn Crisis

The Paradox:

Net Revenue Retention (NRR) of 124.3% suggests the business is healthy. Expansion revenue from existing customers is more than offsetting churn. However, **28.5% of accounts show zero logins for 60+ days** before officially cancelling. This means NRR is being propped up by a shrinking base of highly engaged expanders, while a large silent population is about to churn.

Root Cause Analysis:

1. **Reactive churn management:** We only act when customers cancel, not when they disengage
2. **No early warning system:** No account health score based on engagement velocity
3. **Budget misallocation:** 100% of retention budget goes to expansion, 0% to resurrection
4. **Historical blindness:** 70% of silent churners showed declining engagement for 6 weeks before going dark — we had warning signs

Financial Impact:

- **342 at-risk accounts** representing **\$680K in ARR exposure**
- If silent churn continues at current rate, NRR will drop below 100% within 2 quarters
- CAC payback period increases because we're constantly replacing churned customers instead of retaining them

■ KEY INSIGHT:

Recommended Actions:

1. **Build an 'At-Risk' Playbook:** Trigger interventions at engagement drop (Day 45), not cancellation notice
2. **Launch Re-engagement Campaign:** Target 60+ day inactive accounts with personalized value reminders
3. **Implement Account Health Score:** Login frequency × Feature breadth × Team size = Health Score (0-100)
4. **Reallocate Budget:** Move 20% of expansion budget to resurrection campaigns
5. **Expected Impact:** Save \$680K ARR, reduce silent churn to <15% within 6 months

■ INSIGHT 2: Feature Adoption Breadth Predicts Expansion Revenue

The Discovery:

Accounts using **4+ features generate 3x more expansion revenue** than single-feature users. The correlation is clear: feature breadth = wallet share. However, **70% of users never discover feature #3** (Integrations and Automation). The problem is not product value — it is **feature discovery**.

Root Cause Analysis:

1. **Onboarding focuses on first action only:** Users learn messaging, but never see Integrations or Automation
2. **No forced multi-feature exposure:** Users can complete onboarding without touching 70% of the product
3. **TTV to Feature #3 is 7.2 days:** By then, 60% of free users have already churned
4. **Enterprise features (API, SSO) have lowest adoption but highest expansion correlation (3.5x):** We're not surfacing high-value features

Financial Impact:

- **25% expansion revenue opportunity** is locked behind feature discovery
- API Access and SSO adoption at 22% and 35% respectively — these are enterprise upsell triggers
- Current onboarding loses users before they see multi-feature value

■ KEY INSIGHT:

Recommended Actions:

1. **Redesign Onboarding:** Force multi-feature exposure in first 7 days (messaging → file share → integration)
2. **New Primary KPI:** Measure TTV to Feature #3, not just first action
3. **Feature Discovery Prompts:** Contextual nudges at Day 3 and Day 5 based on usage patterns
4. **Integration Marketplace:** Highlight for Team plan users — show what's possible
5. **Expected Impact:** +25% expansion revenue, 50% increase in 4+ feature adoption

■ INSIGHT 3: Support Tickets Are a Leading Churn Indicator

The Warning:

After a major UI redesign in May 2026, support ticket volume per \$100K MRR increased **25%**. However, CSAT remained flat. This is critical — **users are confused but not angry (yet)**. 53% of tickets are 'confusion' type vs. 15% bugs and 32% feature requests.

Root Cause Analysis:

1. **Redesign improved aesthetics but degraded discoverability:** Users can't find familiar features
2. **No contextual guidance:** Redesigned features launched without tooltips or walkthroughs
3. **Confusion precedes churn by 90 days:** Historical data shows this pattern — we're in the danger window
4. **Support team is reactive, not preventive:** They're answering the same questions repeatedly instead of fixing the root cause

Financial Impact:

- **Support cost increase:** +25% tickets = +25% support headcount or overtime
- **Churn risk:** Confused users churn at 2x the rate of satisfied users
- **Opportunity cost:** Support team time spent on confusion = time NOT spent on high-value customer success

■ KEY INSIGHT:

Recommended Actions:

1. **Categorize & Target:** Set target: confusion tickets <25% of total (currently 53%)
2. **Usability Sprints:** If confusion >40%, halt new features and fix UX
3. **In-App Contextual Guidance:** Tooltips and task completion checklists for redesigned features
4. **A/B Test:** Old vs new UI for power users — measure task completion time
5. **Expected Impact:** -40% support cost, prevent projected churn spike

4. SOLUTIONS & ACTION PLANS

This section consolidates all recommended actions into an executable framework with owners, timelines, and measurable impact.

INSIGHT	HYPOTHESIS	EXPERIMENT	OWNER	IMPACT	STATUS	TIMELINE
Silent Churn 28.5%	Day-45 intervention reduces churn by 40%	Re-engagement campaign for 60+ day inactive accounts (342 accounts)	Customer Success	+\$680K ARR saved	In Progress	Q3 2026
Feature Discovery Gap (70%)	Forced multi-feature onboarding increases 4+ feature adoption by 50%	Redesign onboarding with Feature #3 mandatory walkthrough	Product Team	+25% Expansion Revenue	Planned	Q3-Q4 2026
UI Confusion Spike (+133%)	Contextual in-app guidance reduces confusion tickets by 60%	Add tooltips + task completion checklists for redesigned features	Design + Support	-40% Support Cost	Urgent	Immediate
Sales Cycle Efficiency	Self-serve demo reduces enterprise cycle by 30%	Launch interactive product demo for enterprise prospects	Sales	-12 days Avg Cycle	Completed	Done
Account Health Scoring	Health score predicts churn 6 weeks in advance with 80% accuracy	Build ML model: login freq × feature breadth × team size	Data + CS	Proactive retention	Planned	Q4 2026
Budget Reallocation	20% expansion budget shift to resurrection improves net retention	A/B test: expansion vs resurrection spend allocation	Finance + CS	+5pp NRR	Planned	Q3 2026

4.1 Prioritization Framework

Not all actions can be executed simultaneously. Use the **Impact vs Effort** matrix to prioritize:

QUADRANT	ACTIONS	PRIORITY
High Impact, Low Effort	- In-app contextual guidance for redesigned features - Categorize support tickets by type - Launch re-engagement email campaign	DO FIRST
High Impact, High Effort	- Redesign onboarding with multi-feature exposure - Build account health scoring ML model - Reallocate expansion vs resurrection budget	PLAN & RESOURCE
Low Impact, Low Effort	- A/B test old vs new UI for power users - Add integration marketplace highlights	QUICK WINS
Low Impact, High Effort	- Complete UI redesign rollback - Rebuild entire support knowledge base	AVOID / DEFER

4.2 Success Metrics & KPIs to Track

After implementing solutions, monitor these metrics to validate impact:

1. **Silent Churn Reduction:** Target <15% within 6 months (from 28.5%)
2. **Feature #3 Discovery Rate:** Target 50% of users discover within 7 days (from 30%)
3. **Confusion Ticket %:** Target <25% of total tickets (from 53%)
4. **NRR Stability:** Maintain >120% while silent churn decreases
5. **Support Efficiency:** Tickets per \$100K MRR back to <9.5 (from 12.1)

5. CHART-BY-CHART BREAKDOWN

Each chart in the dashboard serves a specific analytical purpose. Here's how to read them and what questions they answer.

5.1 MRR Movement & Composition (Line Chart)

Chart Type: Multi-line chart with stacked areas

What It Shows:

- **Total MRR trajectory** from \$850K (Jan 2024) to \$4.03M (Jun 2026)
- **Composition:** How much growth comes from New MRR vs Expansion MRR
- **Trend:** Steady growth with acceleration in 2025-2026

Key Question It Answers:

'Is our growth sustainable, or are we overly dependent on new customer acquisition?' **Answer:** Expansion MRR is driving 60% of growth — this is healthy and sustainable.

How to Read:

- The white line = Total MRR
- Blue area = New MRR (new customers)
- Green area = Expansion MRR (existing customers upgrading)
- **Healthy signal:** Both lines trending up, expansion contributing significantly

5.2 NRR vs Silent Churn Divergence (Dual-Axis Line Chart)

Chart Type: Dual-axis line chart with dashed trend line

What It Shows:

- **Left axis (green):** NRR climbing from 105% to 124.3% — looks great
- **Right axis (red dashed):** Silent churn climbing from 15% to 28.5% — hidden danger
- **The divergence:** Two lines moving in opposite directions = paradox

Key Question It Answers:

'Is our retention actually healthy, or are we missing a hidden churn problem?' **Answer:** NRR is being propped up by expansion. Silent churn is a ticking time bomb.

Critical Annotation:

The red vertical line at the end marks the critical threshold: **28.5% silent churn**. This is the moment the dashboard triggers an urgent alert.

5.3 Feature Adoption vs Expansion Correlation (Combo Chart)

Chart Type: Bar chart + line overlay (combo chart)

What It Shows:

- **Bars (left axis):** Adoption rate for each feature (Messaging 95%, File Sharing 82%, Integrations 45%, etc.)

- **Line (right axis):** Expansion revenue correlation multiplier for each feature
- **The gap:** High-correlation features (API, Integrations, Automation) have LOW adoption

Key Question It Answers:

'Which features should we prioritize in onboarding to drive expansion?' **Answer:** Integrations (2.8x), Automation (3.1x), and API Access (3.5x) have the highest expansion correlation but lowest adoption. These are the features to force-expose during onboarding.

Color Coding:

- **Green bars:** High adoption (>70%) — these are table stakes
- **Yellow bars:** Medium adoption (35-70%) — opportunity zone
- **Red bars:** Low adoption (<35%) — high-value, underutilized features

5.4 Time-to-Value vs Conversion by Segment (Scatter Plot)

Chart Type: Bubble scatter plot

What It Shows:

- **X-axis:** Time to Value (days) — lower is better
- **Y-axis:** Conversion rate (%) — higher is better
- **Bubble size:** Segment volume (larger = more users)
- **Four segments:** Free, Team, Business, Enterprise

Key Question It Answers:

'Which segment has the biggest optimization opportunity?' **Answer:** Enterprise is the outlier — 4.5 days TTV but 35% conversion. If we reduce Enterprise TTV to 2.5 days (matching Business), conversion could jump to 45%+.

Segment Analysis:

- **Free (grey, large bubble):** 3.2 days TTV, 8% conversion — low value, high volume
- **Team (blue):** 2.1 days TTV, 15% conversion — sweet spot, optimize here first
- **Business (orange):** 1.8 days TTV, 22% conversion — best-in-class, model for others
- **Enterprise (red):** 4.5 days TTV, 35% conversion — high value, biggest TTV problem

5.5 Enterprise Sales Cycle & Win Rate (Combo Chart)

Chart Type: Bar + line combo chart

What It Shows:

- **Bars (yellow):** Average sales cycle in days (trending down from 65 to 42 days)
- **Line (green):** Win rate percentage (trending up from 22% to 40%)
- **Inverse correlation:** As cycle shortens, win rate improves

Key Question It Answers:

'Are our sales process improvements working?' **Answer:** Yes. The streamlined procurement process reduced cycle by 23 days and doubled win rate from 22% to 40%.

5.6 Support Ticket Decomposition (Stacked Bar Chart)

Chart Type: 100% stacked bar chart

What It Shows:

- **Red (top):** Confusion tickets — spiked after May 2026 UI redesign
- **Yellow (middle):** Bug tickets — stable and declining
- **Blue (bottom):** Feature request tickets — steady
- **Red dashed line:** Marks the UI redesign date

Key Question It Answers:

'What type of support problem is growing, and when did it start?' **Answer:** Confusion tickets spiked 133% post-redesign. Bugs are down. This is a UX problem, not a product quality problem.

5.7 Gross Margin Trajectory (Dual-Axis Line Chart)

What It Shows:

- **Green line (left):** Gross margin climbing from 72% to 88%
- **Red dashed line (right):** COGS declining from \$280K to \$215K
- **Story:** Infrastructure efficiency gains are driving margin expansion

5.8 PQL Funnel Efficiency (Horizontal Bar Chart)

What It Shows:

- **Stage 1:** 13,500 free users
- **Stage 2:** 750 PQLs (5.6% conversion)
- **Stage 3:** 450 sales qualified (60% of PQLs)
- **Stage 4:** 280 closed won (62.2% of SQLs)
- **Overall:** 2.07% free-to-paid conversion

Key Question It Answers:

'Where is our funnel leaking?' **Answer:** The biggest drop is Free → PQL (94.4% don't qualify). Focus here for maximum impact.

6. INTERVIEW STRATEGY

This section prepares you to confidently explain your HTML dashboard in any interview scenario, especially when recruiters or hiring managers expect Power BI or Tableau.

6.1 The Opening Pitch (30 Seconds)

Never start with the tool. Start with the problem you solved.

'I built an interactive executive dashboard for a SaaS B2B collaboration tool to surface critical business insights — like silent churn masking healthy NRR, feature adoption gaps bottlenecking expansion revenue, and a post-redesign support crisis that precedes churn by 90 days. It connects KPIs to causal insights and actionable recommendations, not just reporting numbers.'

6.2 When They Ask 'Why Not Power BI?'

Don't defend. Demonstrate strategic thinking. Here are three response frameworks:

Framework A: The 'Intentional Choice' Response

'I chose to build it from scratch to demonstrate full-stack data storytelling — from raw business logic to interactive visualization. I can absolutely work with Power BI, but this shows I understand what happens under the hood. Knowing how to build visualizations from scratch makes me better at pushing Power BI beyond its default templates.'

Framework B: The 'Business Need' Response

'Power BI is excellent for standard reporting. But for this case study, I needed complete control over interactivity, custom visualizations like the scatter plot for TTV vs Conversion, and the ability to embed strategic narratives directly in the UI. Building it in HTML/Chart.js gave me that flexibility — zero licensing costs, fully portable, and version-controlled on GitHub.'

Framework C: The 'Full Pipeline' Response

'I handled the entire pipeline: data modeling and business logic, visualization design, and frontend delivery. This isn't just a dashboard — it's a decision-support system with drill-down simulations, responsive design for mobile, and embedded action trackers. Power BI would have limited my ability to customize the UX and interactivity to this level.'

6.3 What Your HTML Dashboard Proves (That Power BI Doesn't)

WHAT YOU PROVE	HOW YOU DEMONSTRATE IT	INTERVIEW TALKING POINT
Full-Stack Ownership	You built data logic, visual design, AND frontend code	"I own the entire pipeline from business question to delivered insight"
Custom Interactivity	Drill-down onclick handlers, hover tooltips, responsive design	"I can build exactly the UX stakeholders need, not what the tool allows"
Zero Licensing Barriers	GitHub-hosted, browser-native, no Microsoft ecosystem dependencies	"I can scale this without per-user licensing costs"
Version Control & Collaboration	Every change tracked in Git, CI/CD ready	"My work is reproducible, reviewable, and deployable"

Performance Optimization	Lightweight JS, no heavy BI engine overhead	"It loads faster and runs on any device"
Embedded Storytelling	Narrative insights + action items built into the UI	"I don't just report numbers — I connect them to decisions"
Technical Depth	Chart.js configuration, CSS Grid, responsive breakpoints	"I understand how visualization libraries work under the hood"

6.4 Mock Interview Q&A;

Q: 'This looks like a lot of work for a dashboard. Why not just use Power BI?'

A: 'It was intentional. I wanted to prove I can build a complete data product — not just operate software. Power BI is a tool; this is a solution. I can use both, but this shows deeper capabilities.'

Q: 'How would non-technical stakeholders use this?'

A: 'It's designed for executives — color-coded health status, one-click KPI cards, narrative insights built in. No training needed. It runs in any browser, including mobile. In fact, the responsive design means it works better on mobile than most BI tools.'

Q: 'Can this handle real-time data?'

A: 'Currently it uses static data for the portfolio demonstration, but the architecture is API-ready. I can replace the data arrays with fetch() calls to a backend. The visualization layer is completely decoupled from the data source.'

Q: 'What if the company uses Power BI? Would you rebuild this?'

A: 'No — I'd integrate. I can embed custom HTML visualizations inside Power BI using custom visuals, or build a hybrid where Power BI handles standard reporting and this handles the custom interactive pieces. My HTML skills make me MORE valuable to a Power BI team, not less.'

Q: 'How do you ensure data accuracy?'

A: 'In a production environment, I'd implement data validation at ingestion, automated testing for calculation logic, and reconciliation checks against source systems. For this portfolio piece, the formulas are transparent in the code — anyone can audit them.'

Q: 'What's the most important insight from this dashboard?'

A: 'That NRR of 124% is masking a silent churn crisis. 28.5% of accounts are disengaged but still paying. If we don't act, they'll churn in bulk at renewal. The dashboard flags this 6 weeks early — that's the difference between reporting and decision-making.'

7. TECHNICAL DEEP DIVE

This section explains the technology stack, architecture decisions, and design patterns used to build the dashboard.

7.1 Technology Stack

LAYER	TECHNOLOGY	PURPOSE	ALTERNATIVES CONSIDERED
Frontend Framework	Vanilla HTML5 + CSS3	Zero dependencies, maximum control	React, Vue (overkill for single-page dashboard)
Visualization Engine	Chart.js 4.x	Interactive charts with animations, responsive (scaled automatically)	D3.js (steep learning curve), Plotly (heavier bundle)
Date Handling	chartjs-adapter-date-fns	Time-series chart support for Chart.js	Moment.js (deprecated, heavier)
Styling	CSS Custom Properties (Variables)	Theme consistency, dark mode, easy customization	Sass/LESS (unnecessary for this scale)
Layout	CSS Grid + Flexbox	Responsive grid system without framework	Bootstrap (adds bloat), Tailwind (opinionated)
Interactivity	Vanilla JavaScript (ES6)	Event handling, drill-down simulation, dynamic updates	jQuery (deprecated), React state (overkill)
Hosting	GitHub Pages	Free, version-controlled, instant deployment	Netlify, Vercel (also excellent)
Version Control	Git + GitHub	Change tracking, collaboration, portfolio versioning	GitLab, Bitbucket

7.2 Architecture & Design Patterns

A. CSS Custom Properties (Variables) for Theming

The entire dashboard uses CSS variables for colors, enabling instant theme changes:

```
--bg: #0f172a; --card: #1e293b; --accent: #3b82f6; --success: #10b981;
```

Benefit: Change one variable, update the entire dashboard. Supports light/dark mode toggle with one class.

B. CSS Grid for Responsive Layout

The KPI grid uses `grid-template-columns: repeat(4, 1fr)` on desktop, collapsing to 2 columns on tablets and 1 column on mobile. No media query framework needed.

Benefit: Native browser performance, no JavaScript layout calculations.

C. Chart.js Configuration Pattern

Each chart follows a consistent pattern:

- Get canvas context: `document.getElementById('chartId').getContext('2d')`
- Define datasets with explicit colors, borders, and fills
- Configure responsive options with `maintainAspectRatio: false`
- Add custom tooltip callbacks for currency formatting

Benefit: Consistent UX across all charts, easy to maintain and extend.

D. Event-Driven Interactivity

KPI cards use `onclick='showDrill(kpi)'` to simulate Power BI drill-through pages. In a production environment, this would navigate to a detailed analysis page with filters and segment breakdowns.

Benefit: Demonstrates UX thinking — the dashboard isn't static, it's an exploration tool.

E. Semantic HTML Structure

The dashboard uses semantic HTML5 elements: <header>, <section>, <article> for insight panels. This ensures accessibility (screen readers) and SEO.

Benefit: Professional-grade markup that meets WCAG standards.

7.3 Performance Optimizations

- **CDN-loaded libraries:** Chart.js loaded from jsDelivr CDN — cached globally, faster than self-hosted
- **No external fonts:** Uses system font stack ('Segoe UI', system-ui, -apple-system) — zero FOIT/FOUT
- **Lazy rendering:** Charts render on DOM ready, not blocking page load
- **Minimal DOM nodes:** No unnecessary wrapper divs — clean, shallow DOM tree
- **CSS transitions only:** GPU-accelerated hover effects, no JavaScript animation overhead

7.4 Accessibility (A11y) Considerations

- **Color contrast:** All text meets WCAG AA standards (4.5:1 ratio minimum)
- **Semantic headings:** Proper H1 → H2 → H3 hierarchy for screen readers
- **Keyboard navigation:** All interactive elements (KPI cards) are focusable and clickable
- **Status indicators:** Color-coded pills use both color AND text (success/warning/danger) for colorblind users
- **Responsive text:** Fluid typography scales appropriately on all devices

8. APPENDIX — RAW DATA & CALCULATIONS

8.1 MRR Data Table (Monthly Breakdown)

Source data for the MRR Movement chart:

MONTH	TOTAL MRR	NEW MRR	EXPANSION MRR	CHURNED MRR	NET MRR CHANGE
Jan 2024	\$850,000	\$45,000	\$25,000	\$12,000	+\$58,000
Feb 2024	\$920,000	\$48,000	\$28,000	\$13,000	+\$63,000
Mar 2024	\$1,010,000	\$52,000	\$32,000	\$14,000	+\$70,000
Apr 2024	\$1,080,000	\$49,000	\$35,000	\$15,000	+\$69,000
May 2024	\$1,150,000	\$55,000	\$38,000	\$16,000	+\$77,000
Jun 2024	\$1,220,000	\$58,000	\$42,000	\$17,000	+\$83,000
Jul 2024	\$1,300,000	\$62,000	\$45,000	\$18,000	+\$89,000
Aug 2024	\$1,380,000	\$59,000	\$48,000	\$19,000	+\$88,000
Sep 2024	\$1,450,000	\$65,000	\$52,000	\$20,000	+\$97,000
Oct 2024	\$1,520,000	\$68,000	\$55,000	\$21,000	+\$102,000
Nov 2024	\$1,600,000	\$72,000	\$58,000	\$22,000	+\$108,000
Dec 2024	\$1,680,000	\$75,000	\$62,000	\$23,000	+\$114,000
Jan 2025	\$1,780,000	\$80,000	\$65,000	\$24,000	+\$121,000
Feb 2025	\$1,880,000	\$85,000	\$70,000	\$25,000	+\$130,000
Mar 2025	\$1,980,000	\$90,000	\$75,000	\$26,000	+\$139,000
Apr 2025	\$2,080,000	\$88,000	\$80,000	\$27,000	+\$141,000
May 2025	\$2,200,000	\$95,000	\$85,000	\$28,000	+\$152,000
Jun 2025	\$2,320,000	\$100,000	\$90,000	\$29,000	+\$161,000
Jul 2025	\$2,450,000	\$105,000	\$95,000	\$30,000	+\$170,000
Aug 2025	\$2,580,000	\$110,000	\$100,000	\$31,000	+\$179,000
Sep 2025	\$2,720,000	\$115,000	\$105,000	\$32,000	+\$188,000
Oct 2025	\$2,860,000	\$120,000	\$110,000	\$33,000	+\$197,000
Nov 2025	\$3,020,000	\$125,000	\$115,000	\$34,000	+\$206,000
Dec 2025	\$3,200,000	\$130,000	\$120,000	\$35,000	+\$215,000
Jan 2026	\$3,400,000	\$135,000	\$125,000	\$36,000	+\$224,000
Feb 2026	\$3,550,000	\$140,000	\$130,000	\$37,000	+\$233,000
Mar 2026	\$3,700,000	\$145,000	\$135,000	\$38,000	+\$242,000
Apr 2026	\$3,820,000	\$150,000	\$140,000	\$39,000	+\$251,000
May 2026	\$3,920,000	\$155,000	\$145,000	\$40,000	+\$260,000
Jun 2026	\$4,030,000	\$160,000	\$150,000	\$41,000	+\$269,000

8.2 NRR vs Silent Churn Data

Source data for the NRR divergence chart:

MONTH	NRR (%)	SILENT CHURN (%)	AT-RISK ACCOUNTS	AT-RISK ARR
Jan 2024	105.0%	15.0%	180	\$180,000
Jun 2024	110.0%	18.0%	220	\$264,000
Dec 2024	116.0%	21.0%	260	\$353,000
Jun 2025	122.0%	24.0%	300	\$480,000
Dec 2025	124.0%	28.0%	335	\$620,000
Jun 2026	124.3%	28.5%	342	\$680,000

8.3 Feature Adoption & Expansion Correlation

Source data for the Feature Adoption chart:

FEATURE	ADOPTION RATE (%)	EXPANSION CORRELATION (x)	USER TIER	PRIORITY
Messaging	95%	1.0x	All	Table Stakes
File Sharing	82%	1.2x	All	Table Stakes
Integrations	45%	2.8x	Team+	HIGH OPPORTUNITY
Automation	38%	3.1x	Business+	HIGH OPPORTUNITY
Analytics	28%	2.5x	Business+	MEDIUM OPPORTUNITY
Admin Controls	65%	1.5x	Team+	Standard
API Access	22%	3.5x	Enterprise	HIGHEST OPPORTUNITY
SSO	35%	2.2x	Enterprise	HIGH OPPORTUNITY

8.4 Support Ticket Decomposition

Source data for the Support Ticket chart (monthly totals):

MONTH	CONFUSION	BUGS	FEATURE REQ	TOTAL	CONFUSION %
Jan 2025	120	80	250	450	26.7%
Feb 2025	135	75	270	480	28.1%
Mar 2025	160	70	290	520	30.8%

Apr 2025	200	65	315	580	34.5%
May 2025	280	60	310	650	43.1%
Jun 2025	340	55	325	720	47.2%
Jul 2025	310	50	320	680	45.6%
Aug 2025	250	48	322	620	40.3%
Sep 2025	200	45	335	580	34.5%
Oct 2025	170	42	328	540	31.5%
Nov 2025	150	40	320	510	29.4%
Dec 2025	140	38	312	490	28.6%

Note: May-Jun 2025 spike corresponds to UI redesign launch. Confusion tickets peaked at 47.2% of total.

8.5 PQL Funnel Calculation Detail

Complete funnel math with conversion rates at each stage:

STAGE	VOLUME	CONVERSION FROM PREVIOUS	CONVERSION FROM TOP	REVENUE IMPACT
Free Users (Top)	13,500	—	100%	—
↓ PQLs	750	5.6%	5.6%	Qualification filter
↓ Sales Qualified	450	60.0%	3.3%	Sales pipeline entry
↓ Closed Won	280	62.2%	2.07%	\$4.03M MRR base

Optimization Opportunity:

If we improve PQL Rate from 5.6% to 7.0% (through better onboarding):

- New PQLs = $13,500 \times 7.0\% = \mathbf{945 \text{ PQLs}}$ (vs current 750)
- New Closed Won = $945 \times 60.0\% \times 62.2\% = \mathbf{353 \text{ customers}}$ (vs current 280)
- **Incremental revenue:** 73 new customers $\times$ avg \$350/month = **+\$25,550 MRR = +\$306K ARR**

9. QUICK REFERENCE — ONE-PAGE CHEAT SHEET

Use this section as a quick refresher before your interview. All key numbers, formulas, and talking points in one place.

CATEGORY	KEY NUMBER	WHAT IT MEANS	INTERVIEW TALKING POINT
Revenue	\$4.03M MRR	Monthly recurring revenue	Healthy growth at +18% MoM, driven 60% by expansion
Retention	124.3% NRR	Net revenue retention	Looks healthy BUT masks 28.5% silent churn crisis
Churn	28.5% Silent	Zero logins 60+ days	342 accounts, \$680K ARR at risk — leading indicator
Acquisition	5.6% PQL Rate	Free → qualified lead	72% convert to paid; biggest leak is Free → PQL
Onboarding	2.1 days TTV	Time to first value	Improved from 2.9 days; Enterprise still slow at 4.5
Sales	42 days cycle	Enterprise close time	Down from 65 days; win rate doubled to 40%
Margin	88% Gross	Revenue after COGS	Up from 72%; infrastructure efficiency gains
Support	12.1 / \$100K	Tickets per revenue	+25% spike post-redesign; 53% are confusion
Expansion	3x Revenue	4+ feature users	70% never discover feature #3 — discovery problem
Funnel	2.07% overall	Free → paid conversion	Biggest opportunity: improve PQL rate by 1.4pp

10. FINAL WORDS — HOW TO WIN THE INTERVIEW

The Mindset Shift

Your HTML dashboard is not a limitation — it's a **competitive advantage**. Most candidates show Power BI dashboards they built by dragging and dropping. You built a complete data product from scratch. That demonstrates:

- **Business acumen:** You understand what metrics matter and why
- **Analytical thinking:** You connect data to insights to actions
- **Technical depth:** You can build what you design
- **Ownership:** You took a problem from concept to delivery

These are the exact qualities hiring managers want in a data analyst, product analyst, or business intelligence role.

The 3-Minute Interview Walkthrough

Minute 1 — The Hook: Start with the business problem. 'I built this to solve a silent churn crisis hiding behind healthy NRR.'

Minute 2 — The Insight: Walk through ONE chart deeply. 'Look at this divergence — NRR up, silent churn up. That's the paradox.'

Minute 3 — The Solution: End with action. 'I recommended a Day-45 intervention playbook that could save \$680K ARR.'

Then answer questions. Let them dig into the technical details. You'll be ready.

Remember:

'I didn't build this because I don't know Power BI. I built this because I know how to solve business problems — and sometimes the best solution requires building exactly what you need, not using what you're given.'

— END OF GUIDE —

Good luck with your interview. You've got this.